IT23502 - WEB PROGRAMMING

MINI-PROJECT ASSIGNMENT

UNIT IV – ADVANCED FRAMEWORKS & UNIT V – WEB SERVICES

Duration: One Week

Project Title: Design and Development of a Simple Web Application Using Spring Boot

1. Objective

To develop a simple real-world web application of the students' choice using the concepts covered in Units IV and V. Students are expected to explore the technologies, implement the application themselves, and develop a clear understanding of the concepts used.

2. Team Formation

- Each team may consist of a **maximum of 3 students**.
- All team members must actively participate in the development, demonstration and viva.
- Every student must understand the implementation and be able to explain their individual contribution.

3. Project Requirements

Students may choose any simple application, such as a library system, student record system, event registration system, employee management system or any other suitable application.

The application must demonstrate the following core concepts:

A. Spring Boot, MVC and Database Connectivity

- Create and run a Spring Boot application using Spring Tool Suite (STS).
- Implement a simple MVC-based application.
- Connect the application to a database using JPA and Hibernate.
- Use annotations to define entities and perform basic CRUD operations.
- Demonstrate suitable ORM mapping and one ID generation strategy.

B. Hibernate and Spring Framework Concepts

- Implement at least two HQL queries.
- Demonstrate a simple inheritance mapping example.
- Explain the purpose of `SessionFactory`, `Session` and `Transaction`.
- Demonstrate dependency injection and explain the roles of BeanFactory and ApplicationContext.

- Explain AOP and identify one practical application, such as logging.

The concepts that are not directly integrated into the main application may be demonstrated through short examples.

C. REST APIs and Web Services

- Develop basic REST APIs for the selected application.
- Demonstrate at least two HTTP methods, including GET and POST.
- Test the APIs using Postman or an equivalent tool.
- Explain the basic concepts of web services and the difference between REST and SOAP.

D. Introduction to Microservices

- Explain monolithic and microservice architectures and their differences.
- Develop a small second Spring Boot service or demonstrate a basic independent service.
- Run the services locally on different ports and demonstrate a simple REST-based interaction, wherever feasible.
- Explain the purpose of Spring Cloud and identify a relevant feature.

E. Introduction to DevOps

- Use Git and GitHub to maintain the source code.
- Build and run the application using Maven.
- Explain the purpose and advantages of DevOps and identify relevant tools.

4. Submission Requirements

Each team must submit:

1. Project report containing the problem statement, objectives, design and implementation details.
2. Source code and GitHub repository link.
3. Database schema and sample records.
4. Screenshots of the working application and REST API testing.
5. A simple architecture diagram.
6. Instructions to run the application locally.

5. Submission Instructions

- Upload the project report, source code and related files to the **given Google Drive link** within one week.
- Follow the prescribed file-naming conventions.

Recommended naming format:

- Report: `TeamNo_ProjectTitle_Report.pdf`
- Source code: `TeamNo_ProjectTitle_SourceCode.zip OR github link`

- Presentation: `TeamNo_ProjectTitle_Presentation.pptx`

Include the names and register numbers of all team members in the report.

6. Demonstration and Viva

Project demonstrations and viva will be conducted on:

- **21 October 2026**
- **23 October 2026**

Each team must demonstrate its working application and explain the concepts implemented. All team members must be prepared to answer individual questions about their code, design decisions and the relevant topics from Units IV and V.

7. Evaluation

The mini-project will be evaluated based on:

- Working application and implementation – 30 marks
- Coverage and understanding of Unit IV concepts – 20 marks
- REST APIs and understanding of Unit V concepts – 20 marks
- Code quality, testing and documentation – 10 marks
- Demonstration and individual viva – 20 marks

Total: 100 marks

8. Important Instructions

- Select a simple application that can be completed within one week.
- Implement the project yourselves and understand every component of your code.
- Start with the basic working application and add features only after the fundamentals are working.
- Do not include technologies or features that you cannot explain during the viva.
- You may refer to documentation and learning resources, but you must understand, test and be able to explain the code you submit.
- Marks will be awarded for your implementation, understanding, problem-solving approach and individual contribution, not merely for the number of technologies used.

The main objective is to learn by doing. A small, working application that you understand thoroughly is better than a complex application that you cannot explain.